DO NOW — RECALL

# Database and SQL retrieval challenge

Name: _____ Class: _____ Date: _____

**1. What is a primary key?**

**2. What does INNER JOIN return?**

**3. What is a foreign key?**

**4. A 3×2 zero-indexed array: specify row index range, column range and cell count.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Uniquely identifies a row.

**2. Model answer**

Matched records according to join condition.

**3. Model answer**

Column referencing a key in a related table.

**4. Model answer**

Rows 0–2, columns 0–1, total six cells.

**Vocabulary — know these terms**

primary key — Unique row identifier.

foreign key — Related table key reference.

INNER JOIN — Returns matching joined rows.

WHERE — Filters query results.





# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

**Guided 1 — original CHECK**

Model and check

ID5 is absent from joined result.

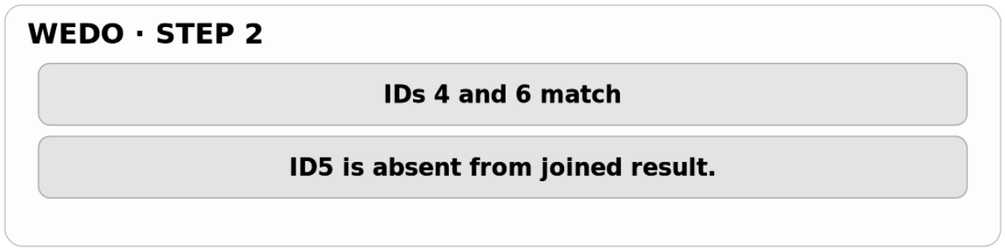
IMPROVE: Explain one correction to your own reasoning.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Student IDs 4,5,6 but Mark IDs 4,6. What appears after INNER JOIN?**

IDs 4 and 6 match




WEDO · STEP 2





NEW LEARNING

# Read, annotate and explain

### Understand

Use a primary key for each Student row and a matching foreign key in Score.

### Trace

INNER JOIN produces one result for each matching StudentID; Cleo has no matching Score and is absent.

### Evaluate

A WHERE clause can filter matched records; forgetting the join condition can produce unwanted combinations.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

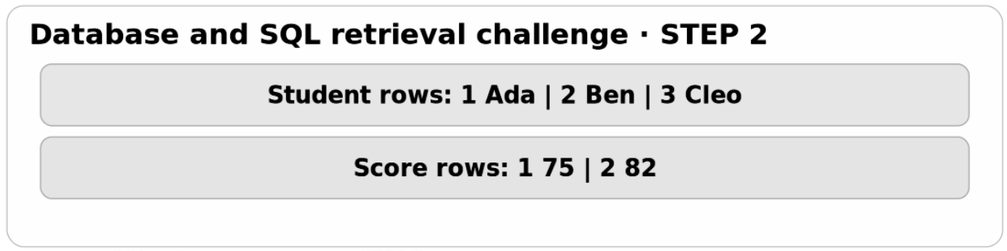
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



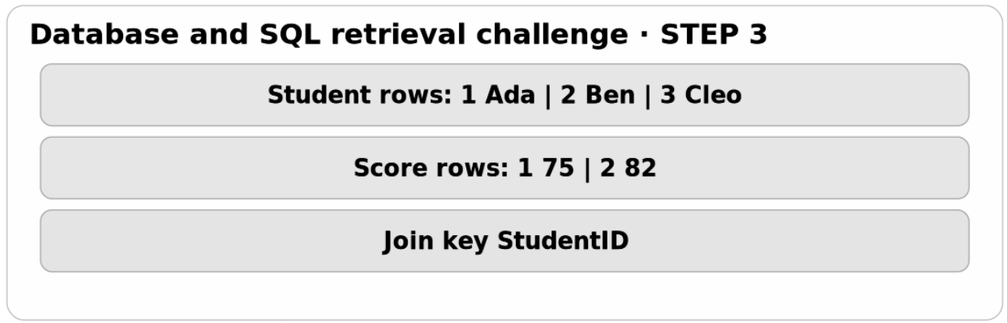

Database and SQL retrieval challenge · STEP 1

**New Learning**




Database and SQL retrieval challenge · STEP 2

**New Learning**




Database and SQL retrieval challenge · STEP 3





**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**INNER JOIN automatically includes rows without a key match.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Explain why omitting a join condition may produce misleading results. [6]

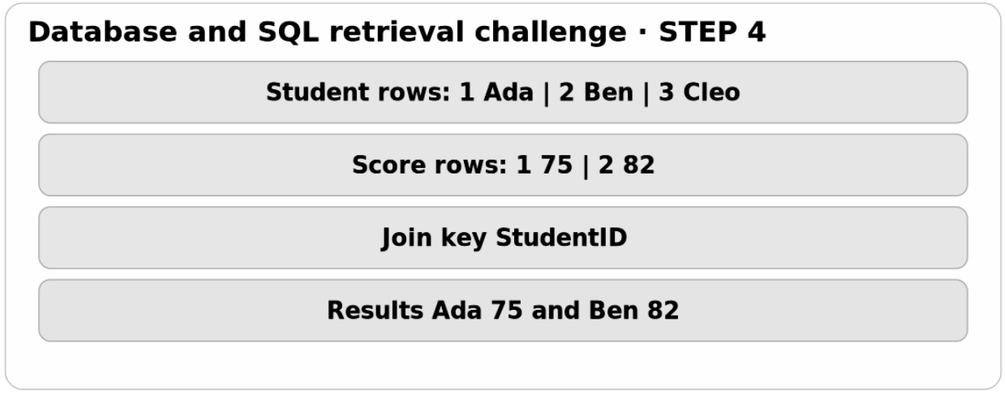




# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Database and SQL retrieval challenge · STEP 4

I DO — WORKED EXAMPLE

# Follow the worked method

**Match Tutor and Class records on TutorID.**

**Predict first**

Tutor 1 matches two class rows

**Model and check**

Tutor 2 has no class and is excluded from INNER JOIN.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

Define primary key, foreign key, INNER JOIN and WHERE. [4]

**Question 2**

Write an INNER JOIN for Student and Score above selecting Name and Mark; give output. [6]

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first IDs 4 and 6 match

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.





# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Match Tutor and Class records on TutorID.**

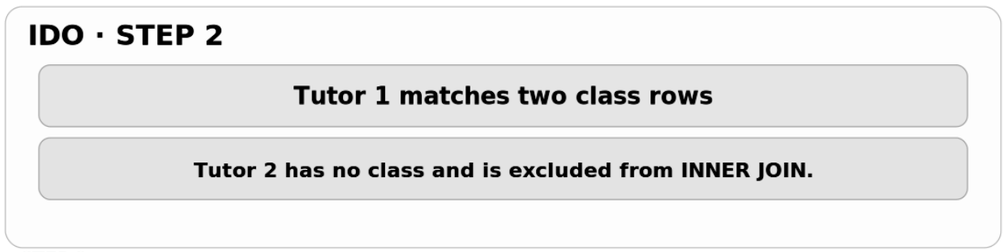
Tutor 1 matches two class rows




IDO · STEP 1

**Match Tutor and Class records on TutorID.**

Tutor 1 matches two class rows




IDO · STEP 2

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve





## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Student IDs 4,5,6 but Mark IDs 4,6. What appears after INNER JOIN?**

IDs 4 and 6 match




WEDO · STEP 1